# Truthprint: Typed Invariant Contracts for Authenticated Semantic Provenance Across Machine Translation

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***Abstract*—Token-level LLM watermarks break under translation and paraphrasing because their signal lives in tokens. We propose *Truthprint*, a semantic intermediate representation that separates *locked invariants* (polarity, quantity, time, attribution, roles) from *controlled realization* via a machine-checkable typed mutability contract. What survives transformation is *ledger-based authenticated semantic provenance*—a keyed tag over the invariant digest, re-verified afterwards—not an in-text watermark: hiding a payload in realization freedom is a negative result under real MT (a surface parser reads $1$ of $4{,}800$ NLLB outputs). On $4{,}800$ NLLB translations across six languages, a Stage-2 extractor recovers the invariants ($0.83$–$0.99$) and the scheme attributes translated documents (core6 document attribution $0.90$–$1.00$), where a token watermark (KGW) and a SemStamp-style embedding-region detector largely fail, and rejects re-translated single-field edits of contract fields ($0.998$; $0.989$ after LLM rewording), which robust hashing does not. Benign rule-based paraphrase authenticates ($1.00$); a fixed-lexicon adaptive attacker removes the mark, while two small open-vocabulary neural runs resist a held-out attack (ValidRemoval $0.05$/$0.10$, $N$=40/10). We state two standard reductions to the MAC and show that the operative false-positive floor is the contract-collision rate, not $2^{-\tau}$. All results are closed-domain and small-scale; Truthprint verifies consistency with a participating keyed generator, not arbitrary AI text.**

***Index Terms*—Intermediate representation, authenticated provenance, LLM watermarking, paraphrase robustness, semantic provenance, SynthID-Text, translation robustness**

## I. Introduction

Provenance for LLM text must survive ordinary transformation. Generation-time watermarking biases decoding so that an authorized detector finds a statistical signature; SynthID-Text integrates this into sampling without retraining and detects it without rerunning the generator [1].

Article 50 of the EU AI Act motivates machine-readable marking [3], useful only if robust [4]; robustness here is a technical objective, not a compliance claim. Provenance also aids data hygiene, since training on unattributed synthetic text degrades models over generations [2].

Token-level methods embed their signal in surface token choices, which translation, paraphrasing, and summarization replace while preserving meaning: the provenance claim concerns content identity, but the carrier depends on language-specific lexical decisions.

Recent semantic watermarks use sentence embeddings, contextual representations, or structured graphs [15], [16], [17], [18], [19], but semantic similarity alone does not distinguish meaning that must never change from choices that may safely vary: two sentences can be embedding-close yet differ in negation, quantity, temporal scope, attribution, modality, or causal direction.

This paper proposes *Truthprint*, a typed semantic IR for provenance that explicitly separates:

1) **semantic invariants**, which must remain fixed; and
2) **controlled realization variables**, which may encode watermark symbols only when semantic and pragmatic equivalence is validated.

The central novelty is a machine-checkable contract that fixes which semantic fields are locked and which realization choices may vary; a keyed tag over the locked fields authenticates meaning out of band—the mechanism that survives real translation, unlike an in-text watermark constrained by the same contract (Sec. X-A). Figure 1 shows the end-to-end data flow.

The delta over the closest prior work is precise. Embedding-based semantic watermarks [15], [16], [17], [19] define admissibility through a *metric* (closeness under a learned distance); Truthprint defines it through a *typed predicate*—admissible only when a symbolic invariant relation holds field by field—so a watermark symbol ranges over key-dependent invariant-equivalence classes rather than tokens or embedding regions (Sec. IX). Conditional on sound extraction, encoding stays within a class and preserves protected fields, and because attribution is gated by a payload bound to the invariant digest, fresh-payload forgery reduces to MAC security. SWAN [18] shares a graph carrier but adopts an existing formalism wholesale, whereas Truthprint types each field with an explicit mutability policy validated before release.

### A. Contributions

- **Method.** A field-level mutability contract (*locked*/*conditional*/*realization-only*) over a typed semantic IR, and ledger-based provenance that registers a keyed tag over the contract digest and re-verifies it after transformation, id-addressed or id-free (Sec. VII). The same contract also constrains an in-text carrier path (keyed realization choices, erasures, error correction), which we report as a negative result under real MT.
- **Conditional analysis.** Assuming sound invariant extraction and validation we establish protected-field fidelity; under standard cryptographic assumptions fresh-payload forgery and per-test tag acceptance reduce to the MAC; and we show

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the operative false-positive floor is the contract-collision rate, with an analytic crossover to the $2^{-\tau}$ bound.
- **Reproducible real-MT evidence.** On $4{,}800$ released NLLB translations across six languages the meaning layer survives and authenticates cross-lingually where surface carriers and a real token watermark do not; at a matched FPR it survives translation at least as well as a SemStamp-style detector and robust hashing (a strong encoder lets SimHash match it) and alone rejects single-field tampers; searched without a document id, it stays specific up to $\sim 10^3$ registered documents but not at $10^5$; and two small open-vocabulary neural runs resist a held-out adaptive attack. All Truthprint numbers reproduce from the released data with no GPU; wide-coverage parsing and scale remain open.

*a) Claim–evidence boundary.:* Table VII (appendix) separates what the artifact *demonstrates*, what is proved only under explicit assumptions, and what remains a hypothesis (e.g., "language-neutral" names the intended schema interface, not a measured property).

## II. Background and Related Work

*a) Token-level and provable watermarks.:* The green–red list watermark [9] and SynthID-Text [1], [5], [6], which generalizes it via tournament sampling, embed a keyed statistic in token selection; analysis shows detector robustness depends on its statistical assumptions [7]. A parallel line seeks formal guarantees: provable robustness for a unigram watermark [10], distortion-free editing-robust watermarks [11], cryptographically undetectable watermarks [12], publicly verifiable unforgeable watermarks [14], and low-FPR statistical tests [13]. All carry the signal in token decisions, which meaning-preserving transformations replace; He et al. show token watermarks lack cross-lingual consistency and are removed by pivot translation [20]. Truthprint inherits the cryptographic spirit at the *provenance* layer—fresh-payload unforgeability and per-candidate tag acceptance (Propositions 1, 2) reduce to MAC security—with no claim of undetectability.

*b) Semantic and structured watermarks.:* Embedding-region and semantic-context methods—SemStamp [15], SemaMark [16], semantic-invariant watermarking [17], dual embeddings [19]—reduce token dependence; SWAN carries the signal in AMR structure [18], and SynGuard hybridizes token masking with semantic alignment [8]. But embedding geometry does not expose *which* typed fact changed: high similarity can coexist with altered polarity, quantity, attribution, modality, or causal direction. Truthprint instead defines a watermark-specific typed IR, assigns each field a mutability policy, and validates every realization against an immutable contract before release.

*c) Attacks and robust hashing.:* Watermark presence and family can be identified by black-box probing [22], motivating a Kerckhoffs threat model (security in the key, not obscurity); scrubbing and spoofing under distillation [24] and radioactive traces in fine-tuned models [23] motivate MAC-authenticated payloads rather than statistical agreement; and detector-guided adversarial paraphrasing [25] motivates counting a removal as successful only when it both evades detection and preserves meaning:

$$\begin{aligned}\text{ValidRemoval} = \mathbf{1}\big[D_K(y') < \tau_D\big] \\ \cdot\ \mathbf{1}\big[\text{InvariantEq}(P(y), P(y')) = 1\big]. \quad (1)\end{aligned}$$

Retrieval over stored generations already defends against paraphrase [29]; Truthprint stores a typed record instead of the text, adding field-level tamper rejection (Sec. X-F). Registering a meaning-digest tag and re-verifying it after transformation also resembles *robust hashing* (MinHash [27], SimHash [28], learned semantic hashing), but those answer a *similarity* question that cannot detect a typed field change—a single-field edit stays within tolerance (Sec. X-E). Truthprint instead hashes a *typed invariant contract* with a keyed MAC, a typed, keyed member of that family whose false-positive floor is the analyzed contract-collision probability (Sec. X-D). Table IX (appendix) positions it against representative systems.

## III. Problem Definition

For a content specification $x$, secret key $K$, payload $m$, encoder $E_K$, language-$\ell$ realizer $R_\ell$, transformation $A$, multilingual parser $P$, and detector $D_K$, the generated text is $y = R_\ell(E_K(x, m))$, an observed transform is $y' = A(y)$, and detection returns $(s, \hat{m}, c) = D_K(P(y'))$ (score, recovered payload or prefix, confidence report). The design targets four objectives: (i) *semantic fidelity* $I(P(y)) = I(x)$ for invariant extractor $I$; (ii) *natural-language quality* above deployment thresholds (only an automatic perplexity proxy here); (iii) *robust detection* under transformations that preserve sufficient invariant and carrier evidence; and (iv) *controlled authenticated false positives*—under the stated PRF and independence assumptions an independently produced decoded payload rarely passes the keyed tag check, with the statistical fallback separately calibrated. In the evaluated ledger setting (Sec. VII) there is no payload $m$: $D_K$ re-extracts $P(y')$ and checks it against registered tags, so (iii) and (iv) concern invariant rather than carrier evidence.

## IV. Threat Model

*a) Benign transformations.:* The target is meaning-preserving edits (normalization, translation, synonym substitution, voice alternation, clause splitting, paraphrase); this paper evaluates translation, round-trip translation, and sentence-level paraphrase.

*b) Attackers.:* A non-adaptive attacker applies off-the-shelf translation, paraphrasing, or summarization without knowing $K$. An adaptive attacker may know the schema and detector design, repeatedly rewrite text, splice documents, target known carrier families, or query an exposed detector.

*c) Out of scope.:* Truthprint does not identify all AI-generated text. It verifies consistency with a participating keyed generator. Absence of a signal does not prove human authorship.

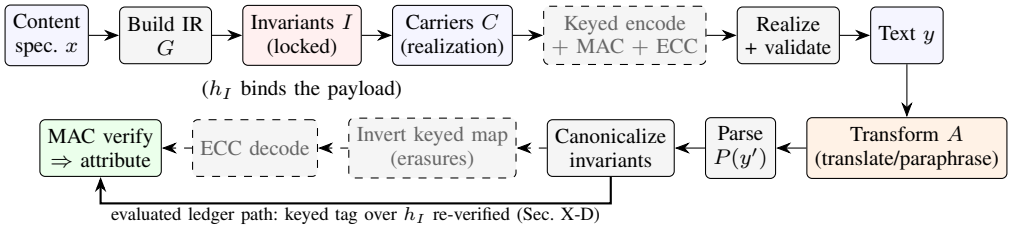



Fig. 1. Truthprint data flow. Encoding (top) locks the invariant set $I$ and validates every candidate realization against the contract before emitting text; detection (bottom) re-parses possibly transformed text and canonicalizes the invariants. Dashed boxes form the in-text carrier path (keyed encoding, inversion, ECC), a negative result under real MT (Sec. X-A). The evaluated mechanism is the solid bypass: a keyed tag over the invariant digest $h_I$, registered out of band and re-verified after re-extraction (Sec. X-D); an invariant-altering transform breaks it.

## V. Truthprint Intermediate Representation

### A. Three-Layer Structure

A Truthprint document contains three conceptual layers.

**Layer A: Semantic invariants.** Entities, predicates, semantic roles, polarity, negation scope, modality, quantities, temporal intervals, causal direction, conditions, exceptions, attribution, quotation status, and uncertainty.

**Layer B: Controlled realization variables.** Voice, event/state framing, clause packaging, discourse-connective family, proposition ordering, nominal/verbal realization, anaphoric choice, and optional explanatory apposition.

**Layer C: Provenance metadata.** Scheme and key identifiers, payload, nonce, ECC parameters, carrier assignments, and invariant digest.

### B. Typed Mutability

Each field $f$ is assigned $\mathrm{Mutability}(f) \in \{\mathrm{locked}, \mathrm{conditional}, \mathrm{realization\text{-}only}\}$: locked fields never change, conditional fields change only when context-specific validators approve, and realization-only fields may carry watermark symbols.

### C. Example Representation

A simplified representation (locked predicate/roles/polarity/-time with a realization space over voice and event framing) is given in Appendix J0b.

## VI. Formal Semantic Contract

A Truthprint graph is $G = (V_E, V_V, E_R, E_D, \Phi)$—entity nodes, event/state nodes, semantic-role edges, discourse/logical relations, and scoped features such as negation, modality, time, quantity, and attribution—partitioned as $G = I \cup C \cup U$ into the invariant set $I$, the controlled carrier set $C$, and unconstrained style $U$. A transformed structure $G'$ is admissible only when $\mathrm{InvariantEq}(I(G), I(G')) = 1$ and $Q(R_\ell(G')) \geq \tau_Q$ for a quality and contextual-consistency function $Q$. Hard invariants are validated symbolically; soft ones (emphasis, implicature) would need entailment checks.

## VII. Ledger-Based Semantic Provenance

This is the mechanism we evaluate. For sentence $i$ with extracted invariants $I_i$ and a declared *contract* $\mathcal{C}$ (the subset of fields that is protected), the producer registers in an out-of-band ledger the truncated keyed tag

$$t_i = \mathrm{MAC}_K\big(h_{\mathcal{C}}(I_i) \parallel n \parallel id_i\big)_{1..\tau}, \tag{2}$$

where $h_{\mathcal{C}}$ hashes the canonical encoding of the contract fields only. A verifier holding $K$ re-extracts $\hat{I}_i$ from the possibly transformed text and accepts iff $\hat{I}_i$ reproduces $t_i$; in this *id-addressed* mode the entry to test is supplied out of band (e.g., by the claimed source). A document is attributed when at least half of its sentences verify. Without an identifier (translation strips in-text ids) the verifier searches the ledger and attributes the document to every registered document that at least half of its sentences match (*id-free*). An edit to a contract field changes $h_{\mathcal{C}}$ and is rejected whenever the extractor reads it; an edit outside the contract passes by design, so the contract fixes the fidelity/robustness trade-off. The document rule tolerates edits to a minority of sentences, so a partial tamper is visible only in the per-sentence verdicts, which the verifier also returns. The key matters only if the ledger is exposed: an unkeyed digest of a low-entropy contract could be inverted by enumeration and new entries minted, whereas a tag needs $K$ (Proposition 1); with a private, trusted ledger a plain digest decides identically.

## VIII. In-Text Carrier Path

The design also lets realization choices carry a payload inside the text: an authenticated, error-corrected payload bound to $h_I$ selects, through keyed permutations, among invariant-equivalent realizations (voice, clause packaging, connective family, proposition order); every candidate is re-validated against the hard invariants and an unrecoverable carrier is recorded as an erasure. Detection re-parses, inverts the keyed map, decodes, and accepts only a payload whose MAC verifies, with a calibrated statistical fallback. This path does *not* survive real translation (Sec. X-A); Appendix J gives the full construction and Appendix J0b its pseudocode.

## IX. Formal Analysis

We state what the design guarantees given sound primitives; what depends on semantic parsing is measured empirically (Sec. X), and computational complexity is in Appendix J.

### A. Carriers as Invariant-Equivalence Classes

**Definition 1** (Invariant equivalence)**.** *Let $I(\cdot)$ be the invariant extractor. Two structures $G, G'$ are invariant-equivalent, written $G \sim_I G'$, iff* $\mathrm{InvariantEq}(I(G), I(G')) = 1$. *For a discourse unit with invariants $I$, the admissible realization set is*

$$\mathcal{R}(I) = \{\, r : I(P(r)) = I \ \wedge\ Q(r) \geq \tau_Q \,\}. \tag{3}$$

A carrier alphabet $A_j$ is a key-dependent partition of $\mathcal{R}(I)$ into classes; the encoder selects a class by symbol and the realizer emits any member. Thus the watermark ranges over the quotient $\mathcal{R}(I)/\!\sim_I$ rather than over tokens or embedding neighborhoods.

**Assumption 1** (Validator soundness)**.** *The hard-invariant validator $V$ is sound: if $V$ accepts a candidate $r$ for invariants $I$, then $I(P(r)) = I$. (It need not be complete; rejected admissible candidates only reduce capacity, not fidelity.)*

**Remark 1** (Conditional protected-field fidelity; immediate from Assumption 1)**.** *Under Assumption 1, every committed output $y = R_\ell(E_K(x, m))$ satisfies $I(P(y)) = I(x)$.*

*Proof sketch.* Encoding commits a candidate only if $V$ accepts it, and carriers with no accepted candidate are recorded as erasures rather than realized (Sec. VIII). By soundness, acceptance implies invariant equality for that unit; taking the conjunction over committed units gives $I(P(y)) = I(x)$. □

*a) Empirical validator soundness.:* We measure the load-bearing assumption as the validator's false-accept rate on single-field tampers. On the closed domain the typed contract flags every tamper across all six fields (property C1, $1.000$, CI $[0.998, 1.000]$) while retaining attribution across benign paraphrase ($0.990$); a decode-only ablation accepts every tamper (C2, $0.000$), confirming that invariant re-extraction, not the coding layer, supplies soundness. Under real NLLB translation of edited text, core6 falsely accepts $0.002$ of single-field edits to its fields (Sec. X-D). Wide-coverage validator soundness remains open, so Remark 1 stays conditional.

### B. Unforgeability and False Positives

**Proposition 1** (Fresh-payload unforgeability; standard reduction)**.** *Model the tag as $t = \mathrm{MAC}_K(x)$ truncated to $\tau$ bits, with $x = h_{\mathcal{C}}(I) \parallel n \parallel id$ for a ledger entry (Eq.* (2)*) or $x = m \parallel n \parallel h_I$ for an in-text payload. Any PPT adversary that causes authenticated attribution for an input $x$ not previously tagged by the generator yields an EUF-CMA forgery against this truncated MAC with the same success probability (up to simulation overhead). Thus its success is bounded by the EUF-CMA advantage of the instantiated truncated MAC; guessing loss is already part of that advantage and is not added a second time.*

*Proof sketch.* Attribution requires a tag equal to $\mathrm{MAC}_K(x)$ on an input not tagged by the generator. The reduction answers authorized generation requests with its tagging oracle and, when the detector accepts a fresh input, outputs it with its tag as the forgery. Replay of a previously tagged input is excluded by freshness and must instead be handled by deployment policy. □

**Proposition 2** (Authenticated acceptance under an ideal-PRF model)**.** *Assume the truncated MAC is a pseudorandom function on a fresh input and the decoded candidate tag is independent of its output. For a fixed (key, nonce) test, an independently produced candidate payload passes the $\tau$-bit tag check with probability $2^{-\tau}$, up to the PRF distinguishing advantage. For $Q$ tested key–nonce pairs, the union bound is $Q2^{-\tau}$ plus that advantage.*

This bounds only a candidate that reaches tag verification: for $q$ fresh candidates whose tags are independent of ideal-PRF outputs the union bound is $q2^{-\tau}$ ($2.3 \times 10^{-10}$ at $\tau$=32), *not* an end-to-end detector FPR—parsing, candidate search, replay, carrier dependence, and threshold calibration all affect system-level error, and replayed outputs require a nonce registry. The $0/20000$ observation of Sec. A is consistent with the bound (approx. $95\%$ binomial upper bound $1.5 \times 10^{-4}$). When full recovery fails, the fallback $z$ of Eq. (6) satisfies $\Pr[z \geq t \mid H_0] \leq e^{-t^2/2}$ under independent binary carriers, calibrated in deployment.

Carrier capacity and the minimum reliable text length for the in-text channel are analyzed in Appendix J.

## X. Experiments

*a) Setup.:* All data are *synthetic template facts* in one closed domain (software incidents: four agents, four patients, one predicate—*fix*, with *break* used only as a tamper value—and two to four closed values each for polarity, quantity, time direction, modality, attribution, and causation, so core6 has $192$ possible values); the main corpus is $50$ documents $\times 16$ sentences, translated by NLLB-200-distilled-600M into Korean, Hindi, Chinese, Arabic, and German and round-trip into English ($4{,}800$ outputs). The *Stage-1* frontend parses surface carriers; the *Stage-2* frontend is a hand-built six-language lexicon that extracts the nine invariant fields. A *contract* is the field subset that is authenticated: core6 (polarity, quantity, time, attribution, predicate, agent) or full9 (all nine). Because the extractor and the corpus share one closed vocabulary, these results establish feasibility, not open-domain accuracy. Runs are seeded; rates carry Wilson $95\%$ CIs; implementation, protocol, and hypotheses are in the appendix.

### A. Surface carriers: a negative result under real MT

On the committed run, the Stage-1 *surface* frontend reads all $800$ English source sentences but only $1$ of the $4{,}800$ NLLB translations (parse coverage $0.000$, CI $[0.000, 0.001]$; the one success is a round trip). Hiding a payload in realization freedom therefore does not survive real translation; the bottleneck is wide-coverage parsing, not the coding or cryptographic core (Appendix A).

### B. The meaning layer survives real translation (Stage-2)

A closed-domain lexicon extractor over six languages (English, Korean, Hindi, Chinese, Arabic, German) recovers the nine invariant fields from $4{,}800$ real NLLB translations ($800$ source sentences over $50$ documents $\times$ six conditions). Aggregate recovery (Wilson $95\%$ CI) is $0.83$–$0.99$: the small

closed categoricals (polarity, quantity, attribution, time) survive at $\geq 0.97$, while the open entity and relational fields are lower (0.83–0.92; patient 0.826 weakest). Strict all-nine-field recovery is highest for Hindi/German (0.69/0.66) and lowest for Arabic (0.26; Arabic loses the patient *cache miss*, Appendix H). The meaning layer thus survives MT where the surface carriers of Sec. X-A do not (`scripts/eval_multilingual.py`).

### C. A neural frontend pilot on open wording

A feasibility pilot (16 base sentences, 6 with open wording, in six conditions; 96 records) ran an instruction LLM as a drop-in extractor: on *closed* templates the hand-built lexicon leads (0.970 vs. 0.726 field recovery), but on the open wording the lexicon degrades below the neural frontend (0.481 vs. 0.614), strongest on typed relational fields (predicate 0.896, causation 0.948). Neural extraction is not yet reliable: it recovers all nine fields of a sentence in none of the 96 records (lexicon 0.479). The harness enforces *separate* translation and extraction models (removing the self-consistency confound).

### D. Meaning-based provenance authentication under real translation

With $\tau$=32 and Stage-2 re-extraction (Sec. VII), the core6 contract gives id-addressed document attribution on the 4,800 translations of 1.000 (CI $[0.929, 1.000]$) for German and Hindi, 0.980 round-trip, 0.940 Korean/Chinese, and 0.900 $[0.786, 0.957]$ Arabic over 50 documents (unchanged after correcting a generator plural bug and re-translating; App. H); sentence-level TPR 0.619–0.901, false positives 0.000–0.007. For tampering we edit one field in the text of each source (in-contract or not), re-realize it, and translate it with the same NLLB system, verifying against the *original* registration: core6 falsely accepts 0.002 (CI $[0.001, 0.005]$, $n$=3,000) of in-contract edits, reading the edit back in 0.742, and accepts 0.751 of edits outside it (patient, modality, causation) by design; full9 rejects those too but falsely accepts 0.009, mostly modality and causation edits that translation renders like the original (0.023/0.045). Reworded once by the same-text LLM before translation, in-contract edits pass core6 at 0.011 $[0.007, 0.018]$ (0.008 in English, where rewording may undo the edit; not human-checked). Widening to nine fields raises meaning coverage but lowers TPR (AR 0.256) and collapses attribution for the hardest conditions (AR 0.02, ZH 0.08, RT 0.28) while German/Hindi hold at 0.94—an explicit fidelity/robustness trade-off. Two low-entropy generations sharing a *contract* collide before the tag is tested, so the false-positive floor is the *contract-collision* probability, not $2^{-\tau}$. For the six-field contract this floor is $5.2 \times 10^{-3}$ ($\approx 1/192$), matching the measured 0.000–0.007 and $\sim 2 \times 10^{7}\times$ the $2^{-32}$ bound; nine fields lower it to $\sim 1.8 \times 10^{-4}$, and the $2^{-\tau}$ bound binds only once the entity-vocabulary entropy exceeds $\tau$ bits (Appendix A). Searched *id-free*, sentence-level matching is not specific—a core6 sentence collides with some wrong entry of the 800-sentence ledger in 0.66–0.89 of cases. The document rule restores specificity: searching all 50 registered documents, core6 attributes translated documents to their source at 0.90–1.00 with $0/2{,}450$ false document attributions per condition (95% upper bound $1.6 \times 10^{-3}$ per document pair). Per-pair risk multiplies with ledger size, which we measure by adding up to $10^5$ distractor documents from the corpus generator (only queries need translation): core6 makes no false attribution up to 1,050 documents, but 1–6 per condition at 10,050 and 3–61 at 100,050 ($\approx 5 \times 10^{-6}$ per pair, near the independence prediction of $1.1 \times 10^{-5}$), whereas full9 makes none up to 100,050 documents ($5 \times 10^{6}$ pairs per condition). At $10^5$ documents no contract is both robust and specific: core6 falsely attributes 6–32% of query documents, full9 attributes only 0.02–0.94.

### E. Baseline comparison: token, semantic, and robust-hashing

A KGW token watermark with $\gamma$=0.25, $\delta$=2.0 [9] on GPT-2 generations, translated by the same NLLB system, loses document-level detection from 0.960 clean to 0.000–0.380 across six conditions (Appendix Table XII; cf. [20]). On the *same* text (50 documents reworded by Qwen2.5-0.5B with KGW, SynthID-Text, or no watermark; Appendix Table XIII) both token watermarks fall to 0.000–0.140 after KO/AR/DE translation, while Truthprint attributes 0.580–0.960 with no false attribution—with caveats: Truthprint alone resamples (up to four times) until its record reads back, 13.3% of sentences still fail, and SynthID-Text is scored by its mean $g$-value, not a trained detector (clean TPR only 0.200). Scored on its first sample only—the baselines' budget; the 207 redrawn sentences are scored on their stored first draw—Truthprint attributes 0.160/0.180/0.660 after KO/AR/DE translation: much of its same-text margin comes from resampling, and only German clearly exceeds the token watermarks. The controlled comparison scores every method at a *matched* 1% FPR on the same 800 NLLB source sentences and their translations (300 bootstrapped 16-sentence documents per condition; static multilingual embedding `model2vec`; Table I). Its document score is the fraction of id-addressed sentences that verify, thresholded at the calibrated FPR (1–2 of 16 for the typed digest); Sec. X-D and Table XIII instead require half the sentences, hence their lower attribution. The typed meaning-digest identifies every translated document in all six conditions (1.000); a SemStamp-style LSH detector [15]—our re-implementation, granted ideal generation—reaches 0.020–0.300, embedding SimHash [28] 0.050–0.973, and MinHash [27] and SHA-256 nothing beyond the round trip. With a paraphrase-tuned multilingual encoder [31] SemStamp-style stays at 0.030–0.213, but SimHash survives translation (0.990–1.000): a strong encoder closes the *survival* gap. Near-duplicate templates handicap few-bit regions (on a separate 16-sentence LLM-MT set SemStamp-style reaches 0.377–1.000; Appendix Table XI). The difference that remains is tampering: a single-field meaning-altering edit is rejected by the typed contract (1.000, CI $[0.981, 1.000]$) but by *neither* similarity hash (0.000; 0.005 with the paraphrase encoder)—a near-duplicate fingerprint cannot see which typed field changed—while the exact hash that rejects every tamper also rejects every legitimate translation (Table I).

TABLE I
MATCHED-FPR COMPARISON ON THE 800-SENTENCE NLLB CORPUS: FRACTION OF TRANSLATED 16-SENTENCE DOCUMENTS IDENTIFIED AT A COMMON 1% FPR (300 BOOTSTRAPPED DOCUMENTS PER CONDITION; THEY REUSE SENTENCES, SO ARE NOT INDEPENDENT). RT = ROUND TRIP VIA KOREAN; SEMSTAMP = OUR RE-IMPLEMENTATION; * = PARAPHRASE-TUNED ENCODER, ELSE STATIC. ONLY THE TYPED CONTRACT ALSO REJECTS SINGLE-FIELD TAMPERS (IN TEXT).

| Method | RT | KO | HI | ZH | AR | DE |
|---|---|---|---|---|---|---|
| Typed | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| SemStamp | 0.300 | 0.020 | 0.057 | 0.053 | 0.023 | 0.020 |
| SimHash | 0.973 | 0.293 | 0.240 | 0.773 | 0.050 | 0.450 |
| SemStamp* | 0.213 | 0.110 | 0.090 | 0.060 | 0.030 | 0.053 |
| SimHash* | 0.997 | 0.990 | 1.000 | 1.000 | 1.000 | 1.000 |
| MinHash | 0.737 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| SHA-256 | 0.020 | 0.000 | 0.000 | 0.000 | 0.000 | 0.000 |

### F. Ledger retrieval and NLI baselines

Because the scheme already uses an out-of-band ledger, the natural control is the established retrieval defense [29]: store the source and *retrieve* it after transformation (bag-of-words cosine nearest neighbor), optionally with an NLI check. Under the same ledger and id-addressed verification, at benign acceptance of $0.94/0.92$ (typed digest $1.000$), plain retrieval and a directional-coverage NLI proxy reject essentially no single-field tamper ($0.047/0.061$; $0.186/0.195$ with a real multilingual encoder): a one-field edit stays nearest its own source. Only a *perfect*, unrealizable oracle NLI matches the typed contract's $1.000$ rejection, and needs a semantic model the typed digest does not (Appendix Table XIV).

### G. Paraphrase and adaptive-attack robustness

Rule-based, in-lexicon paraphrase (voice, time reflow, verb/time synonyms, hedge) authenticates at TPR $1.000$ and document attribution $1.000$, and a single-field altering edit is rejected at $1.000$. A schema-aware adaptive attacker that swaps the predicate verb and time expression for *out-of-lexicon* synonyms achieves ValidRemoval (Eq. (1)) $1.000$ under the *closed* lexicon—a negative result we localize to exactly the abstained fields (predicate and time in all cases; attribution when present), with polarity and agent recovered in $100\%$ of attacks. The vulnerability is the fixed lexicon, not the meaning-digest: an *extended-coverage* frontend drives it $1.000 \rightarrow 0.000$ at unchanged benign and tamper rejection, but a *held-out* out-of-inventory attack still evades any fixed lexicon ($1.000$). Two small runs with real instruction LLMs as open-vocabulary parsers cut the held-out ValidRemoval to $0.050$ (GPT-4o-mini, $N{=}40$, CI $[0.014, 0.165]$) and $0.100$ (Llama-3.3-70B, $N{=}10$, CI $[0.018, 0.404]$) at benign and tamper rejection $1.000$, though Llama still loses $0.200$ of the *seen* attacks that the extended lexicon stops, suggesting open-vocabulary parsing as the general defense (Appendix Tables X, XV).

## XI. CONCLUSION

Truthprint separates immutable meaning from controlled realization with a machine-checkable typed contract; what survives translation is a keyed tag over that contract, re-verified out of band—a complement to token-level watermarks. In one closed synthetic domain, it attributes translated documents ($0.90$–$1.00$) and rejects re-translated contract edits; in-text carriers, id-free search beyond $\sim 10^3$ documents, and unresampled free LLM wording remain weak, and open-domain extraction decides whether it generalizes.

## LIMITATIONS

Multilingual semantic parsing remains imperfect. Not every proposition admits multiple safe realization alternatives. Highly constrained text has low carrier capacity. Some apparently stylistic alternations change emphasis or information structure. Translation can omit modality, evidentiality, subjects, or discourse relations. Strong summarization can destroy both content and watermark evidence. Finally, candidate generation and semantic validation add overhead compared with direct token-level methods: on the same-text run the validation loop needs $1.80$ draws per sentence on average and still finds no valid realization for $13.3\%$ of sentences (Appendix Table XIII). A single tag over the contract digest says *whether* a protected field changed, not *which*; naming the field would need per-field tags or a stored record, which we did not evaluate and which increases exposure. Verification also requires the generator's ledger and, because the tag is a MAC, its secret key: a third party cannot check provenance on its own, and public verification would need a digital signature over the digest (with longer tags), which we have not evaluated. The closed-domain adaptive paraphrase of Section X-G shows the sharpest limitation directly: a schema-aware attacker who swaps out-of-lexicon content synonyms fully removes the meaning-digest under the closed lexicon while preserving meaning. An extended-coverage frontend defends that specific attack (ValidRemoval $1.000 \rightarrow 0.000$ without weakening tamper rejection); a held-out out-of-inventory attack still evades any fixed lexicon, but two real open-vocabulary neural runs (GPT-4o-mini $N{=}40$, Llama-3.3-70B $N{=}10$; Table X) resist even the held-out attack ($0.050/0.100$), suggesting that wide-coverage *open-vocabulary* parsing—not a longer lexicon—is the general defense. Those runs are closed-domain and modest in scale; the current artifact evaluates only one small real generator (Qwen2.5-0.5B) and still does not evaluate human judgments, all official baselines, replay defenses, or open-vocabulary adaptive attacks *at larger scale and across domains*. All evaluation data are synthetic template facts in one closed domain, read by a hand-built extractor that shares the corpus vocabulary, and the tamper edits are template re-realizations or one small LLM's rewording of them, not free human edits; the matched-FPR comparison uses our own SemStamp-style re-implementation on template sentences that handicap few-bit embedding regions (with static and paraphrase-tuned encoders); the same-text token-watermark control uses one small generator ($50$ documents), Truthprint's lower KO/AR attribution there ($0.580/0.580$) shows that freer LLM wording strains the extractor, and without resampling its advantage over the token watermarks holds clearly only for German; and id-free ledger attribution was measured against ledgers of up to $10^5$ documents drawn from the same low-entropy template generator, not real text. Consequently it supports feasibility and

closed-domain robustness claims, not comparative robustness or deployment readiness at wide coverage.

## Ethical Considerations

A provenance detector can be misused to accuse authors, censor content, or make automated disciplinary decisions. Truthprint therefore treats a positive result as evidence about a participating keyed generation system, not proof of AI authorship or of misconduct. Deployments should expose the tested scheme and key identifier, use independently calibrated error rates, preserve an appeal path, and avoid adverse decisions based on the detector alone. Absence of a mark is likewise not evidence of human authorship.

The design has dual-use risks: public carrier families can aid removal attacks, while secret or overly detailed payloads can enable tracking. The ledger retains a typed record of every registered generation, and a deterministic tag reveals when two outputs share a contract, so it must be access-controlled and minimized. Payloads should exclude personal identifiers, prompts, and precise timestamps; keys and nonce registries require access control, rotation, and audit. Evaluation text should be licensed or consented, human annotation should be compensated and documented, and language-specific error disparities should be reported rather than hidden in an aggregate score. The present artifact uses synthetic template facts, no personal data, and no human subjects; the real-MT pilot (Secs. X-A–X-D) translates those synthetic facts with a public open-weights model (NLLB-200-distilled-600M) and involves no personal data or human subjects. Any scaled evaluation that adds human invariant or factual-equivalence annotation must use licensed or consented text and compensated, documented annotation.

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## Appendix

To make the guarantees above concrete and reproducible, we provide a self-contained reference implementation (`reference/truthprint_poc.py`, standard library only) of the language-independent core: the canonical invariant digest $h_I$, the authenticated payload $p = m \parallel \mathrm{MAC}_K(\cdot)$, the invariant-bound keyed carrier map, and a systematic GF(2) code with erasure decoding. This deliberately excludes the semantic parser so that the cryptographic and coding properties can be validated in isolation from model-dependent NLP. It exercises

TABLE II
REFERENCE-IMPLEMENTATION DECODE SUCCESS VS. CARRIER ERASURE RATE ($[96, 48]$ GF(2) CODE, 500 TRIALS PER POINT, SEEDED). FINE RESOLUTION NEAR THE CLIFF REVEALS A GRADUAL TRANSITION FROM FULL RECOVERY AT $\epsilon = 0.42$ TO COLLAPSE AT $\epsilon \geq 0.52$, AS PREDICTED BY PROPOSITION 3.

| Carrier erasure $\epsilon$ | Recovery rate |
|---|---|
| 0.10 | 1.000 |
| 0.20 | 1.000 |
| 0.30 | 1.000 |
| 0.40 | 1.000 |
| 0.42 | 0.998 |
| 0.44 | 0.982 |
| 0.45 | 0.968 |
| 0.46 | 0.950 |
| 0.48 | 0.774 |
| 0.50 | 0.290 |
| 0.52 | 0.000 |
| 0.55 | 0.000 |
| 0.60 | 0.000 |

four properties and reports the following verified behavior (message $\mu = 16$ bits, tag $\tau = 32$ bits, code $[N, k] = [96, 48]$):

- **P1 (round-trip recovery).** An invariant-preserving transform that re-realizes carriers and erases $28/96$ sites still yields exact payload recovery and a verifying MAC.
- **P2 (invariant binding).** Flipping a locked field (polarity positive→negative) changes $h_I$, so the MAC fails and the document is *not* attributed, illustrating Remark 1's contract.
- **P3 (cryptographic false positives).** Over $20{,}000$ independent documents, 0 were attributed. This is consistent with Proposition 2; statistically, the observation alone only gives an approximate 95% binomial upper bound of $1.5 \times 10^{-4}$.
- **P4 (no global rule).** For a fixed symbol, $43/96 \approx N/2$ realized options differ between two documents whose invariants differ, confirming the map is bound to $(K, h_I, n)$.

Table II reports decode success versus carrier erasure rate for the implemented $[96, 48]$ code and seeded random-erasure protocol: full recovery ($\geq 0.998$) through $\epsilon = 0.42$, followed by $0.982$ at $\epsilon = 0.44$, $0.774$ at $\epsilon = 0.48$, $0.290$ at $\epsilon = 0.50$, and zero observed recovery at $\epsilon \geq 0.52$. The location and shape are finite-code measurements; only the necessary survivor-count boundary follows from Proposition 3. These values characterize the coding layer, not detection on model-generated text.

### A. Closed-Domain Linguistic Round-Trip

The core PoC operates on abstract carrier bits. To validate the linguistic layer end to end, a second self-contained script (`reference/truthprint_linguistic_demo.py`) realizes an authenticated payload across templated factual sentences using two invariant-preserving carriers—voice (active/-passive) and time-phrase position—then rewrites, re-parses, and detects on actual sentence strings (the running example of Section E). On a 32-sentence document (64 carriers, code $[64, 32]$, $\tau = 20$) it verifies:

- **L1 (invariant fidelity).** Every benign transform leaves the parsed agent, predicate, patient, and polarity unchanged.
- **L2 (round-trip recovery).** With $19/64$ carriers erased by realization-canonicalizing rewrites, the exact payload is recovered and the MAC verifies.
- **L3 (tamper detection).** Negating a single fact changes the document invariant digest, so attribution correctly fails.

This is a Stage-1 closed-domain demonstration, not a general multilingual parser; it establishes that the contract, keyed carriers, and erasure-aware decoder compose correctly once realized as text, and it isolates the remaining open problem as robust wide-coverage semantic parsing (Section B).

### B. Field-Level Challenge Set and Component Ablation

To test—rather than assume—that a *typed* invariant contract catches what an embedding *similarity* judge misses, we extend the closed-domain frontend to six independently protected fields (polarity, quantity, temporal direction, modality, attribution, causation) and generate minimal-pair probes (`reference` is packaged as `truthprint.challenge`; regenerate with `scripts/eval_challenge.py`). For each field we compare a meaning-preserving realization change against a single-field meaning-altering edit, and judge each with three detectors that differ only in the admissibility rule: the full typed+MAC contract, a decode-only ablation that drops the $h_I$-bound MAC, and an embedding-similarity gate (bag-of-words cosine) *calibrated* to reject at most $5\%$ of legitimate paraphrases. On $300$ documents of $24$ facts ($[48, 24]$ code), with $95\%$ confidence intervals (Wilson for rates, bootstrap for cosines) at a fixed seed, we measure:

- **C1 (typed tamper detection).** The typed+MAC contract detects every single-field tamper across all six fields (aggregate $1.000$, CI $[0.998, 1.000]$) while retaining attribution across benign paraphrases ($0.990$, CI $[0.971, 0.997]$).
- **C2 (MAC is load-bearing).** The decode-only ablation detects *no* tamper (aggregate $0.000$, CI $[0.000, 0.002]$): the carriers still recover, so the $h_I$-bound MAC—not the coding layer—is what resists a meaning-altering edit.
- **C3 (similarity cannot separate meaning).** The calibrated embedding gate catches almost no tamper (aggregate $0.012$, CI $[0.008, 0.018]$), because a single-field edit is lexically smaller than a legitimate voice/time reflow: in $48.6\%$ of cases (CI $[0.463, 0.509]$) the tamper is at least as cosine-similar to the source as the paraphrase is, so *no* similarity threshold can separate them.

This isolates the interaction the design relies on—typed invariants supply the decision an embedding metric cannot, and the MAC binding converts that decision into tamper resistance—on real strings. It remains a closed-domain result; the wide-coverage multilingual analogue requires the neural frontend and human annotations specified in Section B and the accompanying data templates.

### C. Research Questions

The original research questions, with their status in this paper, are:

- RQ1: Does Truthprint preserve immutable meaning better than unconstrained paraphrase-based watermarking?
- RQ2: What true-positive rate is achieved at fixed false-positive rates on clean text?

TABLE III
EVALUATION MATRIX: EACH RESEARCH QUESTION MAPPED TO ITS DECISIVE COMPARISON, MANIPULATED FACTOR, AND PRIMARY OUTCOME MEASURE.

| RQ | Decisive comparison | Primary measure |
|---|---|---|
| RQ1 | Truthprint vs. unconstrained paraphrase watermark | Invariant error rate (IER), contradiction rate |
| RQ2 | Truthprint vs. token/embedding baselines, clean text | TPR at $\{1, 0.1, 0.01\}\%$ FPR; ROC-AUC |
| RQ3 | Pre- vs. post-translation (1-hop, round-trip) | Payload BER, partial-recovery rate |
| RQ4 | Sentence- vs. paragraph-level paraphrase | TPR@1%FPR vs. edit distance |
| RQ5 | Watermarked vs. unwatermarked generation | Quality delta, added latency |
| RQ6 | Recovery vs. document length | Authenticated bits recovered |
| RQ7 | Adaptive attacker with vs. without invariant constraint | ValidRemoval rate (Eq. (1)) |

- RQ3: How much signal survives single-hop and round-trip translation?
- RQ4: How robust is Truthprint to sentence- and paragraph-level paraphrasing?
- RQ5: What quality and latency costs are introduced?
- RQ6: How many authenticated payload bits can be recovered as document length increases?
- RQ7: Can an adaptive attacker remove the watermark while preserving all locked invariants?

Table III fixes, for each question, the single manipulated factor and the outcome that would confirm or falsify the corresponding hypothesis, so that results cannot be reported selectively. *Status.* RQ3 is answered for the ledger path on real NLLB output (Secs. X-B–X-D) and negatively for in-text carriers (Sec. X-A), not via payload BER. RQ2 is answered by the matched-FPR and same-text comparisons (Tables I, XIII) after translation, not at $0.1/0.01\%$ FPR. RQ4 and RQ7 (Eq. 1) are measured on the closed domain (Section X-G). RQ5 is answered only by a perplexity proxy on the same-text run (no latency or human evaluation), and RQ1 and RQ6 were *not* measured: there is no human invariant-error judgment and no recovery-vs-length study on real text.

### D. Diagnostic Simulation (Not an End-to-End Benchmark)

A controlled Stage-1 simulator checks the evaluation code paths and illustrates, under explicitly stipulated channel parameters, how signal placement (token vs. meaning layer) reacts to surface replacement and carrier erasure. Because the "baselines" are method-inspired proxies run from a shared harness rather than the published systems, the simulator *must not be used to rank methods* and is not evidence of real-translation performance. To keep this diagnostic from being mistaken for a benchmark, its full protocol and outputs (Tables IV and V) are deferred to Appendix E; the measured results this paper relies on are the real-MT findings of Sections X-A–X-D and the coding-core verification of Section A.

These hypotheses guided the design. H1 is *not supported* for in-text carriers, which fail under real MT (Sec. X-A); its ledger counterpart is supported on the closed-domain template corpus (Table I) but, on free LLM wording without resampling, clearly only for German (Table XIII). H2–H6 are untested on wide-coverage text:

- H1: Truthprint will outperform token-level watermarking under translation because the carrier is reconstructed from language-independent semantics.
- H2: Explicit invariant validation will reduce errors in negation, quantity, time, causality, and attribution.
- H3: Truthprint may underperform token methods on very short text because semantic carrier capacity accumulates across sentences.
- H4: ECC and erasure-aware decoding will improve recovery under parsing errors and partial deletion.
- H5: Strong summarization will remain difficult because it removes propositions and their associated carriers.
- H6: Combining structured carriers with embedding-based fallback evidence will outperform either alone across diverse languages.

Carrier mappings should depend on the secret key, invariant digest, document nonce, and carrier identity. This prevents simple global carrier rules and reduces forgery risk. Exact detector scores should be access controlled or coarsened to reduce oracle attacks. The detector should localize evidence by paragraph and avoid attributing a mixed document to one source. For the ledger path, a verbatim or translated copy of a genuine output re-verifies by design (it attests the content's source, not the copier), and a spliced document whose majority of sentences comes from one source is attributed to it by the document rule; the per-sentence verdicts expose the rest.

The reference API supplies the nonce to the detector out of band; natural-language carrier recovery alone does not discover it. A deployment must therefore define an authenticated nonce-resolution mechanism, such as a bounded time-bucket search plus deployment identifier or a provenance-ledger lookup. Every tested key–nonce pair consumes false-positive budget and must be included in calibration. A ledger should reject duplicate nonces and distinguish exact replay from fresh attribution. Splicing requires finer-grained binding (for example, paragraph identifiers and a document Merkle root); without it, a valid paragraph proves only that the participating generator produced that paragraph, not the provenance of the surrounding document.

Watermarks should support transparency rather than automated punishment. Every detection report should include the tested key and scheme version, text length, number of reliable carriers, calibrated false-positive rate, language and attack assumptions, confidence, and MAC verification status. User identifiers, raw prompts, and exact timestamps should not be embedded.

### E. Deployment Path and Graceful Degradation

Truthprint is designed to slot into a serving stack as a semantic realization pass rather than as a modification of the token sampler. Two integration modes are practical. In

the *inline* mode, the generator emits a draft, the encoder re-realizes admissible discourse units to carry payload symbols, and the validator gates each candidate; although the coding and cryptographic layers are independent of model size (Sec. IX), the unmeasured generation and validation overhead may be substantial. In the *asynchronous* mode, watermarking runs as a post-generation service, which decouples watermark latency from the user-facing generation latency at the cost of a short delay before provenance is finalized.

Because capacity is unevenly distributed across text (Eq. 5), a deployable system must degrade gracefully. Low-capacity spans—code, formulas, legal clauses, and verbatim quotations—admit few invariant-preserving realizations and are therefore poor carriers. For such spans Truthprint records erasures and can optionally fall back to a token-level watermark such as SynthID-Text, yielding a layered scheme intended to combine semantic-layer transformation tolerance with token-layer short-span coverage; both properties require end-to-end evaluation. A deployed detector should report evidence per paragraph, report the number of reliable carriers alongside the calibrated false-positive rate, and abstain rather than guessing when the recovered capacity falls below the minimum reliable length. This makes the system's competence explicit at detection time, which is the behavior an auditor or platform integrator requires.

Consider the fact with locked invariants FIX(agent=developer, patient=server_error), polarity positive, time *previous_day*. Two binary carriers are available: VOICE $\in$ {active, passive} and TIME-POS $\in$ {front, end}. Suppose the keyed map assigns the code symbols so that this unit must realize as passive with the time phrase at the front:

> *"On the previous day, the server error was fixed by the developer."*

A benign transformation—say an English→Korean translation—typically preserves the invariants but may not preserve every carrier. If translation canonicalizes voice (Korean commonly renders this actively), the VOICE carrier is destroyed; the detector, on re-parsing, flags it as *low confidence* and records an *erasure* rather than reading a wrong bit. The TIME-POS carrier and the carriers of neighboring sentences survive. Because the payload is spread across sentences and protected by an error-correcting code, the authenticated payload is still recovered once the surviving carriers exceed the capacity floor of Eq. (5). In contrast, a rewrite that inserts a negation ("was *not* fixed") alters a locked field, changes the invariant digest $h_I$, and correctly fails MAC verification. This example is executed end-to-end on sentence strings in the reference implementation (Section A, properties L1–L3).

This appendix gives the full protocol and outputs of the controlled Stage-1 diagnostic summarized in the main text. It validates the simulator implementation only; it is not a neural or translation benchmark and must not be used to rank the named published systems.

Tables IV and V report simulator outputs for Truthprint *and* four baselines on a shared closed-domain Stage-1 testbed (500 documents, 32 sentences each). To keep the comparison apples-to-apples in the absence of a wide-coverage neural frontend, every method is evaluated under one identical meaning-preserving channel and one identical protocol; the baselines are method-inspired proxies for selected aspects of each published method's signal placement, run from the same harness (`scripts/eval_baselines.py`), and abstract the neural frontend (LM sampler, sentence encoder, AMR parser) in exactly the way the Truthprint Stage-1 numbers do. This diagnostic checks code paths and illustrates hypotheses under explicitly chosen channel parameters. It is not evidence that any named method attains the reported performance on real translation, and it must not be used to rank the methods. End-to-end evaluation with real generators, translation systems, parsers, official baseline implementations, held-out calibration data, and confidence intervals remains future work.

**Shared channel.** A single meaning-preserving transform is characterised by two physical quantities that all methods see identically: the fraction $\tau_{\mathrm{tok}}$ of surface tokens the transform replaces (cross-script translation replaces almost all tokens) and the fraction $\epsilon_{\mathrm{inv}}$ of meaning-layer carriers lost to parse normalisation (the quantity Truthprint's own analysis uses). Each method's survival is then *derived* from these two numbers through its own signal dependence, so the relative ordering is forced by where the signal lives, not tuned per method: a KGW-style token-identity green-list proxy survives at $(1-\tau_{\mathrm{tok}})^2$; an edit-aligned token watermark (DEW) at $(1-\tau_{\mathrm{tok}})$ up to an edit budget then collapses; a sentence-embedding region (SemStamp) flips near the LSH boundary at a rate $\propto \tau_{\mathrm{tok}}$ but is meaning-dominant; and meaning-layer signals (SWAN, Truthprint) are unaffected by $\tau_{\mathrm{tok}}$, losing only the $\epsilon_{\mathrm{inv}}$ parse fraction.

**Protocol.** For each method a decision threshold is calibrated at $1\%$ FPR on null (un-watermarked) documents; TPR is measured at that threshold after the channel; ROC-AUC is computed from watermarked-vs-null score populations. Truthprint is run through its real linguistic codec on actual sentences ($\mu = 12$, $\tau = 20$, $[64, 32]$ GF(2) code); its quality is unigram token overlap (a BLEU-1-style precision without brevity penalty) between the watermarked and an independently realized copy of the same facts. Baseline fluency/quality is not measured because it requires each method's neural generator, absent from the Stage-1 abstraction.

**Reading the tables.** The clean setting saturates because the proxy generators deliberately select matching symbols and provide many carriers. Under the stipulated high-replacement setting, token proxies lose signal by construction while semantic proxies retain it according to the chosen erasure parameter. These outcomes validate the simulator implementation, not H1 and not any comparison among the named published systems. P2 and P3 separately exercise Truthprint's authenticated coding core (Section A); this harness does not evaluate the security properties of the other systems.

This section pins every choice needed to reimplement the language-independent core exactly, so that the reference scripts and the paper agree symbol for symbol. Nothing here is left to interpretation; the two scripts in `reference/` realize precisely these definitions.

### F. Primitive Instantiation

The reference implementation fixes the following concrete choices.

TABLE VI
MAPPING BETWEEN PAPER NOTATION AND REFERENCE-IMPLEMENTATION IDENTIFIERS.

| Symbol | Meaning | Code identifier |
|---|---|---|
| $h_I$ | canonical invariant digest | `canonical_invariant_digest` |
| $p$ | authenticated payload | `authenticate_payload` |
| MAC | tag verification | `verify_payload` |
| ECC | systematic GF(2) code | `GF2Code.encode` |
| decode | erasure decoding | `GF2Code.decode_erasure` |
| $\pi_j$ | keyed carrier offset | `keyed_bit` |
| $a_j^\star$ | realized option | `realize_option` |
| $\hat{c}_j$ | recovered symbol | `recover_symbol` |
| $D_K$ | detector | `detect_document` |

TABLE IV
DIAGNOSTIC CLEAN-CHANNEL SIMULATION (500 DOCS × 32 SENTENCES; THRESHOLD AT 1% FPR)†. DETECTION SATURATES FOR ALL METHODS ON CLEAN TEXT; THE DISCRIMINATING TRANSFORMED SETTING APPEARS IN TABLE V.

| Method | TPR@1%FPR | ROC-AUC | Overlap |
|---|---|---|---|
| KGW proxy | 1.000 | 1.000 | — |
| SemStamp proxy | 1.000 | 1.000 | — |
| SWAN proxy | 1.000 | 1.000 | — |
| Truthprint | 1.000 | 1.000 | 0.84 |

†Controlled simulation only, not a neural or translation benchmark. Overlap (unigram token overlap with an independently realized copy) measured only for Truthprint (real sentences); baseline fluency needs each method's neural generator. Truthprint additionally applies a 20-bit tag check; this is not a measured end-to-end FPR.

TABLE V
DIAGNOSTIC TRANSFORMED-CHANNEL SIMULATION (TPR AT 1% FPR) $(\tau_{\text{tok}}, \epsilon_{\text{inv}})$ PER LANGUAGE PAIR†: EN→KO $(0.92, 0.30)$, EN→HI $(0.95, 0.35)$, KO→EN $(0.90, 0.28)$. OUTPUTS ILLUSTRATE THE STIPULATED CHANNEL EQUATIONS, NOT REAL TRANSLATION; ROWS ARE METHOD PROXIES IN THE SIMULATOR, NOT IMPLEMENTATIONS. REAL-MT COMPARISONS ARE TABLES I AND XII.

| Method | EN→KO | EN→HI | KO→EN |
|---|---|---|---|
| KGW proxy | 0.020 | 0.018 | 0.030 |
| DEW proxy | 0.000 | 0.000 | 0.000 |
| SemStamp proxy | 0.996 | 0.988 | 0.994 |
| SWAN proxy | 1.000 | 1.000 | 1.000 |
| Truthprint | 1.000 | 0.972 | 0.998 |

†Controlled Stage-1 channel; $\tau_{\text{tok}}$ (token replacement) and $\epsilon_{\text{inv}}$ (meaning-carrier erasure) per translation event. Only the Truthprint proxy in this harness exercises authenticated recovery; this table does not evaluate baseline security.

- **Invariant digest.** $h_I = \text{SHA-256}(\text{JSON}(I))$, where JSON is canonical: keys sorted, separators "`,`" and "`:`", UTF-8 encoded. Canonicalization makes $h_I$ order- and whitespace-independent.
- **Authentication.** MAC = HMAC-SHA256; the tag is the first $\tau$ bits of the MAC over $m \parallel n \parallel h_I$. The payload is $p = m \parallel \text{tag}$ (length $|p| = \mu + \tau$).
- **Keyed carrier map (binary carriers).** $\pi_j = \text{LSB}\big(\text{HMAC-SHA256}(K, h_I \parallel n \parallel \text{ID}_j)\big)$ with $\text{ID}_j$ the 4-byte big-endian carrier index. The realized option is $a_j^\star = \pi_j \oplus c_j$ and the recovered symbol is $\hat{c}_j = \pi_j \oplus a_j^{obs}$.
- **Error-correcting code.** A systematic $[N, k]$ linear code over GF(2) with generator $G = [\, I_k \mid P \,]$, where $P$ is drawn once from a fixed pseudo-random seed. Erasure decoding solves the surviving linear equations by Gaussian elimination over GF(2); it succeeds iff the retained rows have rank $k$ (Proposition 3).
- **Detection decision.** Attribute iff erasure decoding returns some $\hat{p}$ *and* MAC verifies on $(\hat{m}, n, h_I)$; otherwise report the calibrated statistic $z$ of Eq. (6) against a threshold $\tau_z$ without asserting authenticated attribution.

### G. Notation-to-Code Map

Table VI maps paper symbols to identifiers in `reference/truthprint_poc.py`, removing any ambiguity between the formalism and the code.

### H. Generator Plural Fix and Re-translation

An earlier generator pluralized by appending "s", so 153 of the 800 NLLB source sentences read *cache misss*. We fixed the generator, which changes exactly those 153 sources, and re-translated all 800 with the same NLLB system on CPU (`scripts/make_realmt.py`); all 3,882 translations of the 647 unchanged sources are identical to the original GPU run. Every real-MT number in this paper comes from the corrected run. The fix leaves core6 document attribution unchanged in all six conditions and moves sentence-level TPR by at most 0.015 (Hindi $0.886 \rightarrow 0.901$). Scoring the 153 sentences in both runs shows that the patient noun, not the misspelling, drives their behavior: Arabic still loses the patient on the corrected text ($0.007 \rightarrow 0.026$), and core6 recovery is similar ($0.837 \rightarrow 0.843$). These sentences are easier than average for core6 (which excludes the patient), so dropping them lowers Arabic document attribution ($0.900 \rightarrow 0.700$ in both runs). The neural-defense run (Table X) keeps the legacy text because its cached LLM outputs are keyed to it (`paper/results/plural_artifact.md`).

### I. Models, Data, and Licenses

All evaluation text is synthetic and produced by the released code (MIT license). Translation uses NLLB-200-distilled-600M [30] (CC-BY-NC-4.0, non-commercial research use); the token-watermark baselines generate with GPT-2 (MIT) and Qwen2.5-0.5B-Instruct (Apache-2.0), SynthID-Text through the `transformers` implementation (Apache-2.0); the encoders are `paraphrase-multilingual-MiniLM-L12-v2` (Apache-2.0) and `potion-multilingual-128M` (MIT); the neural parsers are GPT-4o-mini (OpenAI API terms) and Llama-3.3-70B-Instruct (Llama 3.3 Community License), queried through OpenAI-compatible APIs. Every use is research evaluation within these terms; because of the NLLB license, the measured pipeline is not itself a deployable commercial system. No model is trained. On four CPU threads (`torch` 2.14.1, `transformers` 5.20.2) rebuilding the 4,800 NLLB outputs takes about 30 minutes (`scripts/make_realmt.py`), the edited-text tamper corpus about 22, its LLM rewording about 30 for all 50 documents, and the same-text run about 2.3 hours for 50 documents; every

scoring script finishes in minutes. The original NLLB and KGW runs (Kaggle GPU) and the neural-parser runs (hosted APIs) did not record wall-clock time.

### J. Parameters, Determinism, and How to Run

The core script uses $\mu = 16$, $\tau = 32$, $[N, k] = [96, 48]$; the linguistic script uses $\mu = 12$, $\tau = 20$, $[N, k] = [64, 32]$ over 32 sentences with two carriers each. All randomness is seeded: message bits and erasure sampling use fixed seeds, and the code matrix $P$ uses a fixed seed, so every reported number is reproducible. The document nonce $n$ is freshly sampled per run and passed explicitly to the detector; recovery is invariant to its value, so success does not depend on the nonce. Both scripts return a non-zero exit code if any asserted property fails, so they double as regression tests. These two scripts need only Python 3 (standard library); the real-MT scoring scripts also need the `truthprint` package, and the rebuild and same-text scripts `torch` and `transformers` (the artifact README lists the commands). The paper builds with `pdflatex` via `make`. The core commands are:

```
python3 reference/truthprint_poc.py             #
    P1-P4
python3 reference/truthprint_linguistic_demo.py #
    L1-L3
make                                            #
    build main.pdf
```

Expected results are those reported in Section A: P1–P4 hold, and L1–L3 hold on sentence strings.

The same primitives are also packaged as an installable Python distribution (`code/` in the accompanying repository) with a unit-test suite and continuous integration. Its command-line interface reruns all properties (`truthprint selftest`) and regenerates Table II deterministically from a fixed seed with 500 trials per erasure rate (`truthprint repro-table`).

This analysis concerns the in-text carrier channel (Sec. VIII, Appendix J), which the body reports as a negative result under real MT (Sec. X-A); it bounds when that channel can work at all.

Model each carrier $j$ as a $q_j$-ary symbol on an erasure channel with erasure probability $\epsilon_j$, where $\epsilon_j$ is the chance that a transformation destroys or obscures that site. The expected recoverable information is

$$\mathbb{E}[\text{bits}] \approx \sum_j (1 - \epsilon_j) \log_2 q_j. \tag{4}$$

With a rate-$R$ code protecting an authenticated payload of $|p| = \mu + \tau$ bits, recovery succeeds with high probability once the surviving capacity exceeds $|p|$ with margin. For $N$ uniform binary carriers ($q_j = 2$) with erasure rate $\epsilon$ and a systematic rate-$R$ code ($k = RN$), this requires

$$(1 - \epsilon)N \gtrsim k \iff \epsilon \lesssim 1 - R. \tag{5}$$

**Proposition 3** (Erasure recovery condition)**.** *For a fixed linear $[N, k]$ code, erasure decoding succeeds exactly when the generator-matrix rows corresponding to the surviving positions have rank $k$. Consequently at least $k$ positions must survive, giving the necessary bound $\epsilon \le 1 - R$ for a realized erasure pattern. Whether recovery occurs near that boundary depends on the code construction and erasure distribution; the reference code's finite-length transition is measured rather than claimed as a universal threshold.*

Equation (5) gives a necessary carrier-count condition $N \ge \lceil (\mu + \tau)/((1 - \epsilon) \log_2 \bar{q}) \rceil$ with $R \le 1 - \epsilon$—a planning bound explaining why short or heavily summarized text falls below capacity (the finite-length transition in Appendix Table II is specific to the implemented code).

Let a document have $S$ discourse units, $|C|$ carriers, $\kappa$ candidates per carrier, and $N$ code positions. Encoding cost is $O(S \cdot |C| \cdot \kappa \cdot c_{\text{val}})$, dominated by the validator cost $c_{\text{val}}$ (a parse plus entailment check). Detection cost is $O(S \cdot c_{\text{parse}} + N \cdot c_{\text{map}} + D_{\text{ECC}})$, where the keyed map $c_{\text{map}}$ is one HMAC evaluation and $D_{\text{ECC}} = O(Nk)$ for GF(2) erasure elimination (worst case $O(Nk^2)$). The cryptographic and coding operations scale with the number of carriers and are independent of the generator parameter count. The evaluated ledger path costs one Stage-2 extraction and one HMAC per sentence, plus one index lookup per sentence when verifying id-free. End-to-end latency of the in-text path is nevertheless unknown because candidate generation, reparsing, and validation dominate and may themselves invoke large models; it must be measured for each implementation.

*a) Encoding.:* The raw payload may carry a generator-family, deployment, coarse time-bucket, nonce, and policy identifier—never personal identifiers, prompts, or exact timestamps. The authenticated payload is $p = m \parallel \text{MAC}_K(m \parallel n \parallel h_I)$ for nonce $n$ and canonical invariant digest $h_I$, error-corrected as $c = \text{ECC}(p)$ following prior robust multi-bit watermarking [21]. The encoder discovers admissible carrier families per discourse unit (voice, clause packaging, discourse-connective family, nominal/verbal realization, proposition order), rejecting context-inappropriate ones such as a passive that obscures an important agent. For carrier $j$ with alternatives $A_j = \{a_{j,0}, \ldots, a_{j,r-1}\}$, a keyed permutation $\pi_j = \text{PRP}_K(\text{CanonicalHash}(I, G, j, n))$ selects $a_j^\star = \pi_j(c_j \bmod |A_j|)$; because the map depends on the invariant graph and nonce, no global rule such as "passive means 1" exists. Each realized candidate is reparsed and checked for hard-invariant equality, bidirectional entailment, numerical and entity consistency, fluency, and expected carrier recoverability; if none is valid the carrier is recorded as an erasure rather than forcing degraded text.

*b) Detection.:* Detection segments the document, identifies language, parses each segment into Truthprint, resolves entities, normalizes predicates and roles, canonicalizes quantities and time, and extracts observable carrier alternatives. For an observed carrier $a_j^{obs}$ it reconstructs the keyed permutation and derives $\hat{c}_j = \pi_j^{-1}(a_j^{obs})$, treating uncertain observations as erasures; an ECC decoder reconstructs candidate payloads, accepted only when their MAC verifies. When complete recovery is impossible, a calibrated fallback over $n$ reliable binary carriers with $k$ keyed matches uses

$$z = \frac{k - 0.5n}{\sqrt{0.25n}}, \tag{6}$$

whose production calibration must account for dependence among carrier observations.

Listing 1. Truthprint encoding pseudocode.

```
G <- BuildTruthprint(x)
I <- ExtractInvariants(G)
p <- AuthenticatePayload(K, m, nonce, Hash(I))
c <- ErrorCorrectingEncode(p)
C <- DiscoverAdmissibleCarriers(G)
for assignment in KeyedAssign(K, I, C, c):
    G2 <- ApplyCarrier(G, assignment)
    candidates <- RealizeCandidates(G2, language)
    accepted <- ValidateAgainstContract(candidates,
     I)
    if accepted is empty:
        MarkErasure(assignment)
    else:
        CommitBestCandidate(accepted)
return AssembleDocument()
```

Listing 2. Truthprint detection pseudocode.

```
segments <- SegmentAndIdentifyLanguage(text)
observations <- []
for segment in segments:
    G, conf <- ParseTruthprint(segment)
    I <- CanonicalizeInvariants(G)
    for carrier in ExtractObservableCarriers(G):
        if Reliability(carrier, conf) >= threshold:
            observations.append(InvertKeyedMap(K, I
    , carrier))
        else:
            observations.append(ERASURE)
candidates <- ECCDecode(observations)
valid <- VerifyMACs(K, candidates)
score <- CalibratedStatisticalScore(observations, K
    )
return score, Best(valid), ConfidenceReport()
```

A prototype can use the following modular layout:

```
truthprint/
  schema/        # entities, events, relations,
    mutability
  parser/        # multilingual parser and
    canonicalizer
  carriers/      # voice, packaging, discourse,
    ordering
  encoding/      # payload, keyed mapping, ECC
  realization/   # candidate generation and ranking
  validation/    # hard constraints, NLI, numbers,
    quality
  detection/     # extraction, decoding, statistics
  attacks/       # translation, paraphrase, summary
    , splice
  experiments/   # datasets, metrics, runners
```

## K. Core Data Model

Listing 3. Illustrative Python data model.

```
from enum import Enum
from pydantic import BaseModel, Field

class Mutability(str, Enum):
    LOCKED = "locked"
    CONDITIONAL = "conditional"
    REALIZATION_ONLY = "realization_only"

class Event(BaseModel):
    id: str
    predicate: str
    roles: dict[str, str]
    polarity: str = "positive"
    modality: str = "asserted"
    time: dict = Field(default_factory=dict)
    quantity: dict | None = None
    attribution: str | None = None

class CarrierSite(BaseModel):
    carrier_id: str
    family: str
    target_node_ids: list[str]
    alternatives: list[str]
    recoverability: float
    quality_cost: float
```

## L. Keyed Mapping Skeleton

Listing 4. Secret-keyed carrier assignment skeleton.

```
import hmac
from hashlib import sha256

def keyed_index(key, invariant_hash, nonce,
                carrier_id, symbol, option_count):
    msg = b"|".join([
        invariant_hash,
        nonce,
        carrier_id.encode(),
        symbol.to_bytes(4, "big")
    ])
    digest = hmac.new(key, msg, sha256).digest()
    return int.from_bytes(digest[:8], "big") %
    option_count
```

What this paper did and did not evaluate, against the protocol of Section B. Items marked *not evaluated* are open, not implied.

## M. Baselines

*Evaluated:* unwatermarked generation; KGW on GPT-2 (different source text) and on the same Qwen2.5-0.5B text; SynthID-Text (`transformers`, mean $g$-value) on the same text; our SemStamp-style re-implementation; embedding SimHash; MinHash; SHA-256; ledger retrieval alone and with an NLI proxy or oracle. *Not evaluated:* official SemStamp, SemaMark, semantic-invariant watermarking, SWAN, DEW, a trained SynthID detector, and a standardized toolkit such as MarkLLM [26].

## N. Languages and Domains

*Evaluated:* English source with NLLB translation into Korean, Hindi, Chinese, Arabic, and German, plus a Korean round trip; one closed synthetic domain (software incidents). *Not evaluated:* news, encyclopedic, technical, scientific, policy, dialogue, and creative text, and low-capacity spans (code, formulas, legal clauses, quotations).

## O. Attacks

*Evaluated:* direct and round-trip translation; rule-based benign paraphrase; single-field meaning-altering edits, applied to the record, to the source text, and to an LLM rewording of it before NLLB translation; a closed-lexicon adaptive paraphrase and a held-out out-of-inventory attack (Section X-G). *Not evaluated:* free-form human edits, Unicode and spelling edits, two-hop translation, deletion, insertion, splicing, summarization, regeneration, and detector-guided rewriting.

TABLE VII
CLAIM–EVIDENCE MAP FOR THE PRESENT SUBMISSION (PART 1 OF 2: METHOD AND ROBUSTNESS CLAIMS; PART 2 IS TABLE VIII). "NOT ESTABLISHED" MEANS THAT THE CLAIM MUST NOT BE INFERRED FROM THE DIAGNOSTIC SIMULATION.

| Claim | Status | Supporting evidence or missing test |
|---|---|---|
| Coding/authentication round trip | Demonstrated | Deterministic core tests P1–P4 and closed-domain string tests L1–L3. |
| Protected-field fidelity | Conditional (validator soundness measured) | Remark 1, assuming a sound invariant extractor and validator. Soundness is measured as the false-accept rate on meaning-altering edits on the closed domain (0 observed, C1); under real MT, edited source text re-translated by NLLB is falsely accepted at $0.002$ (core6, in-contract fields) and $0.009$ (full9) for template edits, and at $0.011$ (core6) when a small LLM rewords the edit first (Sec. X-D); wide-coverage soundness remains open. |
| Fresh-payload unforgeability | Conditional | Reduction to EUF-CMA security of the MAC; replay and nonce discovery are separate deployment concerns. |
| Surface-carrier translation robustness | Refuted (real MT) | Stage-1 surface parser reads $800/800$ source sentences but $1/4{,}800$ real NLLB translations (Sec. X-A). |
| Meaning-layer recovery under real MT | Demonstrated (closed-domain, six languages; data committed) | Stage-2 extractor recovers fields at $0.83$–$0.99$ over $4{,}800$ real NLLB translations in six languages (closed categoricals $\geq 0.97$; open entity/relational fields $0.83$–$0.92$), and the meaning-digest authenticates documents cross-lingually (core6 document attribution $1.000$ for German/Hindi, $0.90$–$0.98$ for the rest; Secs. X-B, X-D). The full run ships in-repo (`paper/results/realmt_b2/`); wide-coverage parsing and larger document counts remain open. |
| Neural parsing on open wording | Pilot (positive per field; no sentence fully exact) | In a small LLM-MT pilot (16 base sentences, 6 with open wording; 96 records) a neural extractor beats the lexicon on open wording ($0.614$ vs. $0.481$ field recovery) but recovers all nine fields exactly in none of the 96 records (lexicon $0.479$) while the lexicon leads on closed templates; single-model and small-scale caveats apply (Sec. X-C). |
| Benign paraphrase robustness | Demonstrated (closed-domain) | On 200 documents $\times 16$ sentences, meaning-preserving in-lexicon paraphrase (voice, time reflow, verb/time synonyms, discourse hedge) authenticates at TPR $1.000$ and document attribution $1.000$; single-field altering edits rejected $1.000$ (Sec. X-G). |
| Adaptive-attack robustness | Closed lexicon refuted; defended by extended coverage; open-vocabulary open | A schema-aware attacker swapping out-of-lexicon content synonyms ("took care of" for "fixed") reaches ValidRemoval $1.000$ (Eq. 1) under the closed lexicon, localized to the stripped predicate/time/attribution fields (polarity and agent survive $100\%$). An extended-coverage frontend recovers those fields, driving adaptive ValidRemoval to $0.000$ while benign auth ($1.000$) and tamper rejection ($1.000$) are unchanged; a held-out out-of-inventory attack still evades any fixed lexicon (ValidRemoval $1.000$), but two small real open-vocabulary neural runs (GPT-4o-mini $N{=}40$, Llama-3.3-70B $N{=}10$) resist even the held-out attack (ValidRemoval $0.050/0.100$, CIs disjoint from the lexicons') at benign/tamper $1.000$. Coverage, not the meaning-digest, is the lever; the neural frontend supplies open-vocabulary coverage (Sec. X-G, Table X). |

### P. Metrics

*Reported:* TPR at a matched $1\%$ FPR, ROC-AUC (same-text), document attribution, sentence authentication, tamper rejection, per-field recovery, false attributions per ledger pair, ValidRemoval, and a perplexity fluency proxy (same-text). *Not reported:* TPR at $0.1/0.01\%$ FPR, payload bit-error rate on real text, minimum reliable length on real text, latency, entailment-based integrity, and human quality or factual-equivalence judgments.

The invariant error rate of Table III (not measured here) is

$$\mathrm{IER} = \frac{\sum_i w_i E_i}{\sum_i w_i}, \tag{7}$$

where critical categories such as polarity and numeric value receive larger weights or are treated as hard failures.

TABLE VIII
CLAIM–EVIDENCE MAP (PART 2 OF 2, CONTINUED FROM TABLE VII): FALSE-POSITIVE AND BASELINE-COMPARISON CLAIMS.

| Claim | Status | Supporting evidence or missing test |
|---|---|---|
| System-level false-positive rate | Not established | The tag bound is per tested fresh candidate; parser search, tested key–nonce pairs, and fallback scores require end-to-end calibration. |
| Semantic-level false-positive rate (contract collision) | Measured (closed-domain) | The $2^{-\tau}$ bound covers cryptographic tag collision only; the meaning-digest floor is the contract-collision probability. On the closed-domain corpus this floor is $5.2 \times 10^{-3}$ ($\approx 1/192$) for the six-field contract—matching the measured $0.000$–$0.007$ real-MT false positives and $\sim 2 \times 10^{7}\times$ the $2^{-32}$ bound (Sec. X-D). Open-domain text has lower contract entropy, so estimating it per corpus is a deployment prerequisite. |
| Comparative robustness vs. token and semantic baselines | Demonstrated, small scale (real MT: KGW token; matched-FPR SemStamp-style + robust hashing) | Under the same NLLB translator a KGW watermark on GPT-2 generations collapses (TPR@1%FPR $0.960$ clean $\rightarrow$ $0.000$–$0.380$ translated) while meaning-digest document attribution on the template corpus stays at $0.90$–$1.00$; different source texts, so not a controlled head-to-head (Table XII, Sec. X-E). At a matched $1\%$ FPR on the $800$-sentence NLLB corpus the typed digest identifies every translated document ($1.000$, six conditions) while a SemStamp-style re-implementation reaches $0.020$–$0.300$ (templates handicap embedding methods; $0.377$–$1.000$ on a separate 16-sentence set); with a paraphrase-tuned encoder SimHash also survives ($0.990$–$1.000$), so survival alone does not separate the methods, and only the typed contract rejects single-field tampers (Table I, Sec. X-E). On the same LLM text (Table XIII) KGW and SynthID-Text fall to $0.000$–$0.140$ after translation, Truthprint attributes $0.580$–$0.960$ with resampling and $0.160$–$0.660$ on a single sample ($50$ documents, one $0.5$B generator); SWAN and a trained SynthID detector remain future work. |
| Distinction from robust/semantic hashing | Demonstrated (closed-domain, real MT) | Robust hashes (exact, MinHash, SimHash) either die under translation or cannot detect a typed field change (reject $\leq 0.005$ of single-field tampers vs. typed $1.000$); the meaning-digest is a typed, keyed member of the family with an analyzed collision floor (Secs. II-0c, X-E). |
| Distinction from store-and-retrieve (ledger) baseline | Demonstrated (closed-domain) | Under the same ledger, retrieval and a cheap NLI proxy reject $\leq 0.06$ of single-field tampers at matched benign acceptance ($\leq 0.20$ with a real multilingual encoder); only a perfect (unrealizable) oracle NLI reaches $1.000$, and needs a semantic model, which the typed digest does not (Sec. X-F). |
| Translator-independent (separated) frontend | Pilot (open-domain positive on one field) | A model-separated embedding frontend recovers open-domain temporal direction ($0.944$) where the lexicon abstains ($0.000$); static embeddings remain insufficient for the full contract, motivating the separated-backend LLM frontend (Sec. X-C). |

TABLE IX
POSITIONING OF TRUTHPRINT RELATIVE TO REPRESENTATIVE WATERMARKING AND ATTACK RESEARCH.

| Category | Representative work | Main carrier or focus | Limitation addressed by Truthprint |
|---|---|---|---|
| Token-level sampling | SynthID-Text [1] | Keyed token-selection statistics | Translation and paraphrasing replace token decisions. |
| Sentence-space watermarking | SemStamp [15] | Keyed sentence-embedding regions | Embedding regions are not typed by invariant category. |
| Semantic-context watermarking | SIR [17], SemaMark [16] | Contextual semantic embeddings | Similarity does not explicitly protect negation, numbers, causality, or attribution. |
| Hybrid defense | SynGuard [8] | Token watermark plus semantic alignment | Semantic constraints are less explicit than a typed IR contract. |
| Structured semantics | SWAN [18] | AMR-level structures | Existing semantic form is not a complete watermark-specific mutability contract. |
| Dual embeddings | DEW [19] | Context and token embeddings | Language invariance remains implicit rather than represented as a canonical IR. |
| Attack analysis | SynthID-Text analysis [7], black-box probing [22], adversarial paraphrasing [25] | Detector assumptions and adaptive transformation | Motivates authenticated payloads, conservative interfaces, and erasure-aware decoding. |
| Proposed system | Truthprint | Typed invariants and controlled realization classes | Explicitly specifies what must remain fixed and what may carry symbols. |

TABLE X

OPEN-VOCABULARY NEURAL FRONTEND VS. FIXED LEXICONS ON THE ADAPTIVE CORPUS (CONTRACT CORE6, CACHED REAL-LLM EXTRACTIONS; WILSON 95% CI). *Two* INSTRUCTION LLMS RESIST THE *held-out novel* ATTACK THAT BOTH FIXED LEXICONS FAIL, AT UNCHANGED BENIGN AUTH AND TAMPER REJECTION. $N$ = ITEMS SCORED PER FRONTEND; THE ADAPTIVE-SEEN COLUMN SCORES THREE ATTACK VARIANTS PER ITEM ($3N$). THE CORPUS KEEPS THE PRE-FIX GENERATOR TEXT (APPENDIX H).

| Frontend | Benign TPR | Adaptive-seen VR | Held-out novel VR | Tamper rej. |
|---|---|---|---|---|
| Closed lexicon ($N$=40) | 1.000 | 1.000 | 1.000 | 1.000 |
| Extended lexicon ($N$=40) | 1.000 | 0.000 | 1.000 | 1.000 |
| Neural: GPT-4o-mini ($N$=40) | 1.000 | 0.050 [0.023, 0.105] | 0.050 [0.014, 0.165] | 1.000 |
| Neural: Llama-3.3-70B ($N$=10) | 1.000 | 0.200 [0.095, 0.373] | 0.100 [0.018, 0.404] | 1.000 |

TABLE XI

MATCHED-FPR COMPARISON ON A SEPARATE SET OF 16 BASE SENTENCES TRANSLATED BY AN LLM (NO GERMAN CONDITION); THE 300 DOCUMENTS PER CONDITION RESAMPLE THESE 16 SENTENCES. SAME METHODS AND PROTOCOL AS TABLE I.

| Method | RT | KO | HI | ZH | AR |
|---|---|---|---|---|---|
| Typed digest | 1.000 | 1.000 | 1.000 | 1.000 | 1.000 |
| SemStamp | 1.000 | 0.937 | 0.617 | 0.460 | 0.377 |
| SimHash | 1.000 | 0.590 | 0.000 | 0.883 | 0.633 |
| MinHash | 1.000 | 0.000 | 0.000 | 0.000 | 0.000 |
| SHA-256 | 0.867 | 0.000 | 0.000 | 0.000 | 0.000 |

TABLE XII

TOKEN WATERMARK VS. MEANING DIGEST UNDER THE SAME REAL NLLB TRANSLATOR. KGW (ON GPT-2 GENERATIONS) IS SCORED PER DOCUMENT AT A CALIBRATED 1% FPR; THE TRUTHPRINT ROWS ARE CORE6 ON THE TEMPLATE-FACT CORPUS AT ITS CONTRACT-COLLISION OPERATING POINT ($\sim 5 \times 10^{-3}$). SOURCE TEXTS AND OPERATING POINTS DIFFER, SO THIS SHOWS TOKEN-SIGNAL REMOVAL, NOT A CONTROLLED HEAD-TO-HEAD; THE CLEAN CELL (—) IS UNDEFINED FOR A LEDGER SCHEME WITH NO IN-TEXT SIGNAL. SECTION X-E (TABLE I) GIVES THE MATCHED-FPR COMPARISON WITH A SEMSTAMP-STYLE DETECTOR AND ROBUST HASHING. REPLACES THE DIAGNOSTIC SIMULATOR (APPENDIX E).

| Method | clean | RT | KO | HI | ZH | AR | DE |
|---|---|---|---|---|---|---|---|
| KGW (token) | 0.960 | 0.380 | 0.040 | 0.040 | 0.020 | 0.000 | 0.080 |
| Truthprint, doc. attribution | — | 0.980 | 0.940 | 1.000 | 0.940 | 0.900 | 1.000 |
| Truthprint, sentence TPR | — | 0.720 | 0.684 | 0.901 | 0.666 | 0.619 | 0.850 |

TABLE XIII

SAME-TEXT CONTROL: ONE GENERATOR (`QWEN2.5-0.5B-INSTRUCT`, CPU) REWORDS THE FACTS OF ALL 50 COMMITTED 16-SENTENCE DOCUMENTS UNWATERMARKED, WITH KGW ($\gamma$=0.25, $\delta$=2), AND WITH SYNTHID-TEXT (`TRANSFORMERS` TOURNAMENT SAMPLING, UNWEIGHTED MEAN $g$-SCORE); ALL ARE TRANSLATED BY NLLB-200-DISTILLED-600M. TOKEN WATERMARKS: TPR ABOVE THE MAXIMUM SCORE OF THE 50 UNWATERMARKED DOCUMENTS (EMPIRICAL FPR 0/50) AND ROC-AUC. TRUTHPRINT USES THE UNWATERMARKED TEXT, ACCEPTING A SAMPLE ONLY IF STAGE-2 READS BACK ITS CORE6 RECORD (UP TO 4 SAMPLES; 0.868 [0.842, 0.889] OF SENTENCES VALIDATED, THE REST COUNT AS FAILURES), AND ATTRIBUTES DOCUMENTS ID-FREE AGAINST THE 50 REGISTERED DOCUMENTS. FALSE ATTRIBUTIONS: 0/50 IN EVERY CONDITION. WILSON 95% CI IN BRACKETS. `PAPER/RESULTS/SAME_TEXT_NLLB.MD`. FACT FIDELITY OF THE CLEAN ENGLISH TEXT, BY THE EXTRACTOR-BASED PROXY (CORE6 READ BACK; NOT HUMAN-JUDGED): UNWATERMARKED FIRST SAMPLE 0.609 [0.574, 0.642], KGW 0.504 [0.469, 0.538], SYNTHID-TEXT 0.517 [0.483, 0.552]; THE VALIDATION LOOP REACHES 0.868 AT A MEAN OF 1.80 DRAWS PER SENTENCE (`SAME_TEXT_FIDELITY.MD`). ON ITS FIRST SAMPLE ONLY, TRUTHPRINT'S DOCUMENT ATTRIBUTION IS 0.940 CLEAN AND 0.160/0.180/0.660 FOR KO/AR/DE (`SAME_TEXT_NORESAMPLE.MD`). FLUENCY PROXY (PERPLEXITY UNDER GPT-2, NOT THE GENERATOR): UNWATERMARKED FIRST SAMPLE 95.6, KGW 84.4, SYNTHID-TEXT 107.7, TRUTHPRINT VALIDATED 87.3 (`SAME_TEXT_QUALITY.MD`); NO HUMAN QUALITY JUDGMENT.

| Condition | KGW TPR | KGW AUC | SynthID TPR | SynthID AUC | TP doc. attr. | TP sent. auth. |
|---|---|---|---|---|---|---|
| clean | 0.920 [0.812,0.968] | 0.998 | 0.200 [0.112,0.330] | 0.870 | 1.000 [0.929,1.000] | 0.868 |
| KO | 0.040 [0.011,0.135] | 0.607 | 0.000 [0.000,0.071] | 0.697 | 0.580 [0.442,0.706] | 0.481 |
| AR | 0.040 [0.011,0.135] | 0.545 | 0.140 [0.070,0.262] | 0.533 | 0.580 [0.442,0.706] | 0.470 |
| DE | 0.080 [0.032,0.188] | 0.631 | 0.000 [0.000,0.071] | 0.500 | 0.960 [0.865,0.989] | 0.716 |

TABLE XIV

LEDGER RETRIEVAL(+NLI) VS. THE TYPED MEANING-DIGEST, SAME LEDGER AND ID-ADDRESSED VERIFICATION, BENIGN-PARAPHRASE ACCEPTANCE MATCHED AT $\approx 0.95$ (960 CLOSED-DOMAIN ITEMS, WILSON 95% CI). STORING AND RETRIEVING THE SOURCE SURVIVES PARAPHRASE BUT CANNOT REJECT A SINGLE-FIELD TAMPER; ONLY A PERFECT (UNREALIZABLE) NLI MATCHES THE TYPED CONTRACT.

| Method | Benign TPR | Tamper rej. | Model |
|---|---|---|---|
| Meaning-digest (typed) | 1.000 | 1.000 | none |
| Retrieval (NN) | 0.941 | 0.047 | encoder |
| Retrieval + NLI (proxy) | 0.918 | 0.061 | NLI |
| Retrieval + NLI (oracle) | 0.941 | 1.000 | perfect NLI |

TABLE XV

PARAPHRASE, ADAPTIVE ATTACK, AND EXTENDED-LEXICON DEFENSE ON REAL ENGLISH STRINGS (200 DOCUMENTS $\times$16 SENTENCES, CONTRACT CORE6, WILSON 95% CI). BENIGN MEANING-PRESERVING PARAPHRASE IS AUTHENTICATED (RQ4); A SCHEMA-AWARE ADAPTIVE ATTACKER (RQ7) REMOVES THE MARK UNDER THE *closed* LEXICON (VALIDREMOVAL 1.000), BUT AN *extended-coverage* FRONTEND RECOVERS THE SAME INVARIANTS AND DRIVES VALIDREMOVAL TO 0.000 WHILE STILL REJECTING ALTERING EDITS; A HELD-OUT ATTACK WHOSE SYNONYMS LIE OUTSIDE THE EXTENDED INVENTORY STILL EVADES (COVERAGE BOUND). AUTH TPR IS OVER MEANING-PRESERVING PARAPHRASES; TAMPER REJECTION FOR THE ALTERING ROW.

| Transform family | Auth TPR | ValidRemoval | Surface carrier |
|---|---|---|---|
| Benign paraphrase | 1.000 | 0.000 | 0.015 |
| Adaptive (RQ7), closed lexicon | 0.000 | 1.000 | 0.000 |
| Adaptive (RQ7), extended | 1.000 | 0.000 | 0.000 |
| Held-out novel, extended | 0.000 | 1.000 | 0.000 |
| Altering (tamper), extended | 0.000 | — (1.000 rej.) | 0.000 |